

Math 141, F11

Name: *Answer*

Quiz 1

Section:

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(6 points) Find the domain of the following functions.

(a) $f(x) = \sqrt{x} + \sqrt{10-x}$

Solution: $x \geq 0$ and $10-x \geq 0$
 $\Rightarrow 0 \leq x \leq 10$ (or $[0, 10]$)

(b) $f(x) = \frac{3}{x^2 + 2x}$

Solution: $x^2 + 2x \neq 0$
 $x(x+2) \neq 0$
 $x \neq 0$ and $x \neq -2$
(or $(-\infty, -2) \cup (-2, 0) \cup (0, +\infty)$)

Problem 2.(4 points) Sketch the graph of the function $y = 2^x$ in the following:

